

Association between chemical mixtures and female fertility in women undergoing assisted reproduction in Sweden and Estonia

Bellavia et al.

Supplementary material

Methods

Table S1: List of internal standard bisphenols, phthalate metabolites and parabens

Compound	Abbreviation	Internal standard
Mono-(2-ethyl-5-carboxypentyl) phthalate	MECPP	MECPP-13C4 (ring-1,2 & dicarboxyl)
Mono-(2-ethyl-5-hydroxyhexyl) phthalate	MEHHP	MEHHP-13C4 (ring-1,2 & dicarboxyl)
Mono(2-ethyl-5-oxohexyl) phthalate	MEOHP	MEOHP-13C4 (ring-1,2 & dicarboxyl)
Mono-(2-ethyl-1-hexyl) phthalate	MEHP	MEHP-13C4 (ring-1,2 & dicarboxyl)
Mono-n-octyl phthalate	MnOP	MnOP-13C4 (ring-1,2 & dicarboxyl)
Cyclohexane-1,2-dicarboxyl acid, mono-(7-carboxy-4-methylheptyl ester)	cxMiNCH	cxMiNCH-13C4 (ring-1,2 & dicarboxyl)
Cyclohexane-1,2-dicaboxylic acid mono-(7-hydroxy-4-methyloctyl)ester	ohMiNCH	ohMiNCH-13C4 (ring-1,2 & dicarboxyl)
Cyclohexane-1,2-dicaboxylic acid mono-(4-methyl-7-oxooctyl)ester	oxoMiNCH	oxoMiNCH-13C4 (ring-1,2 & dicarboxyl)
Mono-(4-methyl-7-carboxyheptyl)phthalate	cxMiNP	cxMiNP-13C4 (ring-1,2 & dicarboxyl)
Mono-(4-methyl-7-hydroxyoctyl)phthalate	ohMiNP	ohMiNP-13C4 (ring-1,2 & dicarboxyl)
Monoethyl phthalate	MEP	MEP-13C4 (ring-1,2 & dicarboxyl)
Monocyclohexyl phthalate	MCHP	MCHP-13C4 (ring-1,2 & dicarboxyl)
Monoisobutyl phthalate	MiBP	MiBP-13C4 (ring-1,2 & dicarboxyl)
Mono-n-butyl phthalate	MnBP	MnBP-13C4 (ring-1,2 & dicarboxyl)
Mono(2-hydroxyisobutyl)phthalate	MOHiBP	MOHiBP-d4-3,4,5,6
Mono(3-hydroxybutyl)phthalate	MOHBP	MOHBP-d4-3,4,5,6
Methyl paraben	MePAR	MePAR-d4-2,3,5,6
Ethyl paraben	EtPAR	EtPAR-d4-2,3,5,6
Isopropyl paraben	iPrPAR	iPrPAR-d4-2,3,5,6
Propyl paraben	PrPAR	PrPAR-d4-2,3,5,6

Isobutyl paraben	iBuPAR	iBuPAR-d4-2,3,5,6
Butyl paraben	BuPAR	BuPAR-d4-2,3,5,6
Bisphenol A	BPA	BPA-d18
2,4-Bis(4-hydroxyphenyl)-4-methyl-1-pentene	MBP	MBP-d18
Bisphenol F	BPF	BPF-13C12 (phenoxy)
Bisphenol S	BPS	BPS-13C12 (phenoxy)
4-Hydroxyphenyl 4-Isopropoxyphenyl Sulfone	BSIP	BPSIP d7 (butyl)
Bisphenol AF	BPAF	Bisphenol-AF d4 - 6,6',8,8'
Bisphenol AP	BPAP	BPAP-d5 (phenyl)
Bisphenol B	BPB	Bisphenol-B (phenoxy)d8
Bisphenol E	BPE	BPE-13C6 (phenoxy)
Bisphenol Z	BPZ	Bisphenol-Z (phenoxy)13C12
Bisphenol C2	BPC2	Bisphenol-C2 (phenoxy)13C12

Table S2: List of internal standard PFAS

Compound	Abbreviation	Internal standard
Perfluoro-n-butanoic acid	PFBA	BPFA-13C4
Perfluoro-n-pentanoic acid	PFPeA	PFPeA-13C5
Perfluoro-n-hexanoic acid	PFHxA	PFHxA-13C5 (1,2,3,4,6)
Perfluoro-n-heptanoic acid	PFHpA	PFHpA-13C4 (1,2,3,4)
Perfluoro-n-octanoic acid	PFOA	PFOA-13C8
Perfluoro-n-nonanoic acid	PFNA	PFNA-13C9
Perfluoro-n-decanoic acid	PFDA	PFDA-13C6 (1,2,3,4,5,6)
Perfluoro-n-undecanoic acid	PFUnDA	PFUnDA-13C7 (1,2,3,4,5,6,7)
Perfluoro-n-dodecanoic acid	PFDoDA	PFDoDA-13C2 (1,2)
Perfluoro-n-tridecanoic acid	PFTTrDA	PFDoDA-13C2 (1,2)
Perfluoro-n-tetradecanoic acid	PFTeDA	PFTeDA-13C2 (1,2)
Perfluoro-1-butanedisulfonate	PFBS	PFBS-13C3 (1,2,3)
Perfluoro-1-pentadisulfonate	PFPeS	PFBS-13C3 (1,2,3)
Perfluoro-1-hexadisulfonate	PFHxS	PFHxS-13C3 (1,2,3)
Perfluoro-1-heptadisulfonate	PFHpS	PFHxS-13C3 (1,2,3)
Perfluoro-octadisulfonate (sum)	PFOS	PFOS-13C8
1H,1H,2H,2H Perfluorohexane sulfonate	42FTS	42FTS-13C2 (1,2)

1H,1H,2H,2H Perfluorooctane sulfonate	62FTS	62FTS-13C2 (1,2)
1H,1H,2H,2H Perfluorodecane sulfonate	82FTS	82FTS-13C2 (1,2)
1H,1H,2H,2H Perfluorododecane sulfonate	102FTS	102FTS-13C2-d4 (1,2)
N-Methylperfluorooctanesulfonamidoacetic acid	MeFOSAA	MeFOSAA-d3 (methyl)
N-Ethylperfluorooctanesulfonamidoacetic acid	N-EtFOSAA	N-EtFOSAA-d5 (ethyl)
Perfluoro-1-octanesulfonamide	FOSA	FOSA-13C8
Tetrafluoro-2-(heptafluoropropoxy)propanoic acid	HFPO-DA	HFPO-DA-13C (1,2,3)
11-chloroeicosafuoro-3-oxaundecane-1-sulfonate	11CIPF30UdS	PFDODA-13C2 (1,2)
9-chlorohexadecafluoro-3-oxanonane-1-sulfonate	9CIPF30UdS	PFUnDA-13C7 (1,2,3,4,5,6,7)
Dodecafluoro-3H-4,8-dioxanonanoate	DONA	PFHpA-13C4 (1,2,3,4)

The native and isotopically labelled compounds described in tables A and B were bought from Merck, Toronto Research Chemicals, Cambridge Isotope laboratories, Chiron, CDN Isotopes and Wellington Laboratories.

Applied sample preparation bisphenols, phthalate metabolites and parabens

Internal standard (50 µl, 10 ng/ml of each standard compound) was added to 200 µl follicular fluid. For the hydrolysis of the conjugated metabolites 5 µl β-glucuronidase (0.7 U per sample, Roche) and 0.2 ml ammonium acetate buffer (Merck) (0.5 M, pH 6.2) were added. The deconjugation step was performed for 180 minutes at 37°C and quenched by adding 0.5 ml 5% NH₄OH (Merck) solution.

For the isolation of the deconjugated metabolites SPE was applied using 10 mg Oasis MAX cartridges (Waters). The cartridges were conditioned with 1 ml 2% formic acid (Biosolve) in methanol (Biosolve), 1 ml methanol and 1 ml water. After the sample loading, the cartridges were washed with 1 ml 5% NH₄OH and 1 ml acetonitrile (Biosolve).

The elution was performed with 1 ml 2% formic acid in methanol. The eluates were evaporated till dryness and the residues were reconstituted in 50 µl water/methanol 80/20 v/v mixture. The extracts were stored at 4°C until analysis.

Applied sample preparation PFAS

Prior to the extraction, 50 µl internal standard (10 ng/ml of each standard compound), 0.3 ml 1% formic acid and 0.2 ml methanol was added to 200 µl follicular fluid. The 10 mg Oasis WAX SPE cartridges (Waters) were conditioned with 1 ml 0.1% NH₄OH in methanol, 1 ml methanol and 1 ml water.

Following the loading, the cartridges were washed with 1 ml water and dried by negative pressure.

The 2 ml 0.1% NH₄OH in methanol eluates were evaporated till dryness and the residues were reconstituted in 50 µl water and methanol 50/50 v/v mixture. The extracts were stored at 4°C until analysis.

Performance characteristics bisphenols, phthalate metabolites and parabens

Prior to the analysis of the follicular fluid samples, the recovery and repeatability of the analytical method were determined using fortified bovine follicular fluid at a level of 2 ng/ml (n=8).

In parallel with the samples (n = 333) 22 blanks (MilliQ) were analyzed. The collected blank data were used for the determination of the limits of detection (LODs) and limits of quantification (LOQs). The LOD, the lowest analyte concentration likely to be reliably distinguished, is defined as three times the standard deviation of the blank plus the average blank value. The LOQ, the lowest concentration at which the analyte can not only be reliably detected but at which some predefined goals for bias and imprecision are met, is defined as $3.3 * \text{LOD value}$.

The fortified bovine follicular fluid samples were also analyzed as quality control sample included in each LC-MS/MS run. The reproducibility was determined with the values of the quality control samples (n=8 in the course of the analyses).

Performance characteristics PFAS

Similar as for the bisphenols, phthalate metabolites and parabens, the recovery and repeatability of the analytical method was determined using fortified bovine follicular fluid, at a level of 5 ng/ml (n=8).

In parallel with the follicular fluid samples from Estonia (n=148) 11 blanks (MilliQ) were analyzed. The blank data were used for the determination of the limits of detection (LODs) and limits of quantification (LOQs). The LOD is defined as three times the standard deviation of the blank plus the average blank value. The LOQ is defined as $3.3 * \text{LOD value}$.

The fortified bovine follicular fluid samples were also analyzed as quality control sample included in each LC-MS/MS run. The reproducibility was determined with the values of the quality control samples (n=7 in the course of the analyses).

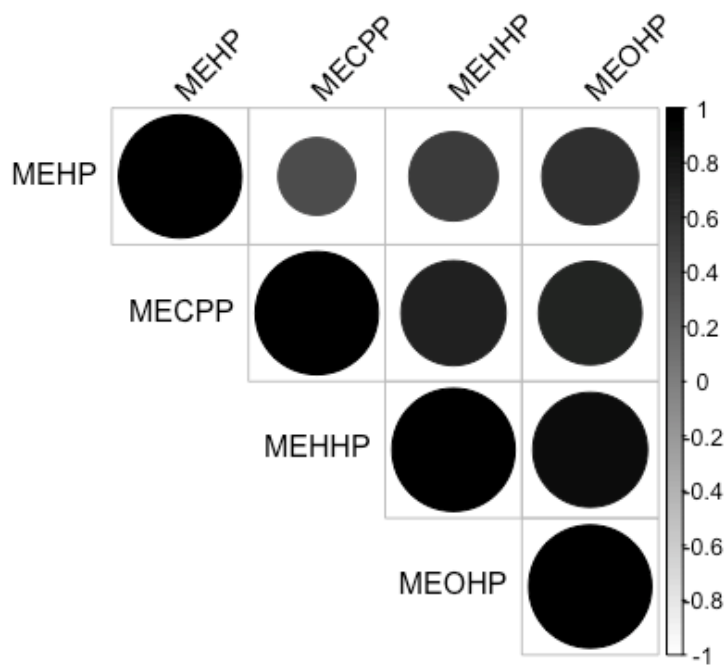
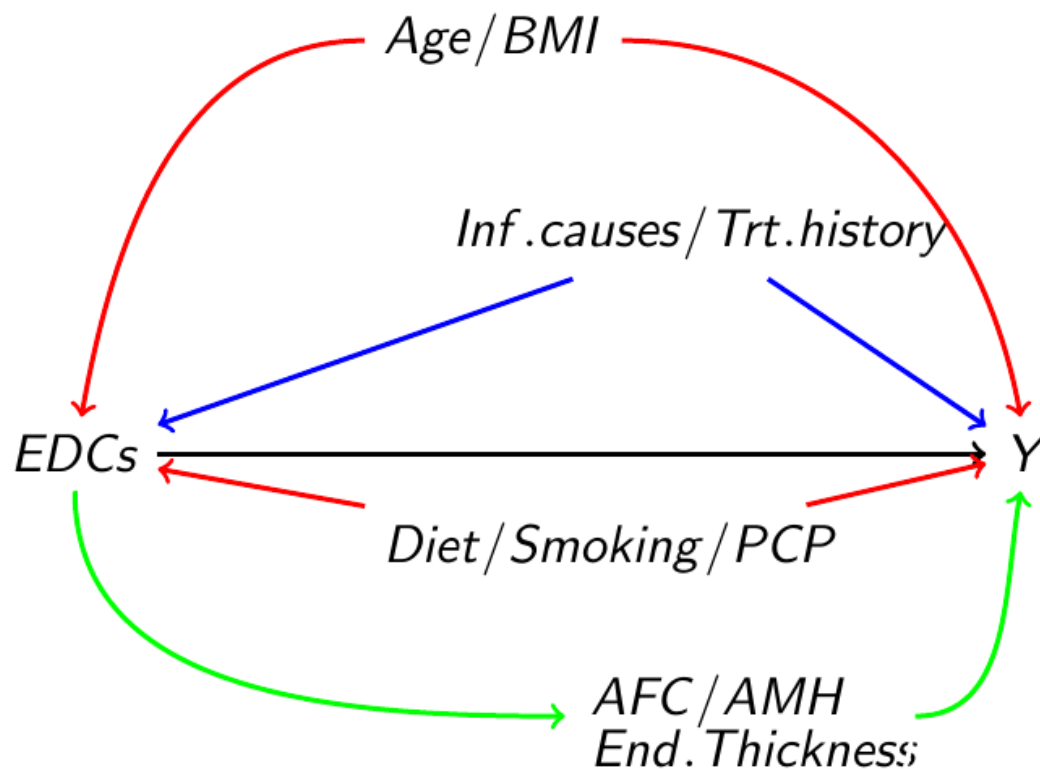
Figure S1. Correlation plot of DEHP metabolites

Figure S2. Direct acyclic graph describing the potential effect of measured covariates on the association between EDCs and outcomes Y (black arrow). Patient characteristics such as age, BMI, and lifestyle factors (diet, smoking, personal care products use) are potential confounders of the associations (red arrows). Covariates related to infertility causes and treatment (Trt.) history are also potential confounders (blue arrows). Indicators such as AFC/AMH and endometrial (End.) thickness are instead potential mediators of the association (green arrows) as it is more plausible to assume that they are affected by EDCs exposure rather than affecting it.



Results

Table S3 includes the distributions and levels of quantification of all evaluated chemicals, including the ones presented in the manuscript in Table 3.

Table S3. Observed distributions and levels of quantification of evaluated chemicals (ng/mL)

	Sweden (n=185)	Estonia (n=148)	Combined population (n=333)	Level of Quantification
PFHxS				0.007
Mean (SD)	1.15 (1.69)	0.231 (0.227)	0.735 (1.34)	
Median [Min, Max]	0.595 [0.0137, 13.0]	0.180 [0.0400, 2.40]	0.320 [0.0137, 13.0]	
Non-quantified	5 (2.7%)	1 (0.7%)	6 (1.8%)	
PFOA				0.1
Mean (SD)	1.35 (0.900)	0.726 (0.398)	1.07 (0.783)	
Median [Min, Max]	1.16 [0.169, 8.05]	0.630 [0.190, 3.20]	0.850 [0.169, 8.05]	
Non-quantified	4 (2.2%)	1 (0.7%)	5 (1.5%)	
PFOS				0.01
Mean (SD)	4.09 (2.43)	2.42 (1.66)	3.34 (2.27)	
Median [Min, Max]	3.54 [0.171, 15.1]	1.90 [0.520, 12.0]	2.68 [0.171, 15.1]	
Non-quantified	4 (2.2%)	2 (1.4%)	6 (1.8%)	
PFNA				0.007
Mean (SD)	0.581 (0.348)	0.455 (0.330)	0.524 (0.345)	
Median [Min, Max]	0.506 [0.0204, 3.16]	0.370 [0.0800, 2.30]	0.441 [0.0204, 3.16]	
Non-quantified	4 (2.2%)	1 (0.7%)	5 (1.5%)	
PFHpA				0.007
Mean (SD)	0.0373 (0.0482)	0.0217 (0.0171)	0.0299 (0.0378)	
Median [Min, Max]	0.0247 [0.0105, 0.485]	0.0200 [0.00700, 0.115]	0.0200 [0.00700, 0.485]	
Non-quantified	41 (22.2%)	20 (13.5%)	61 (18.3%)	
PFDA				0.001
Mean (SD)	0.284 (0.152)	0.183 (0.145)	0.239 (0.157)	
Median [Min, Max]	0.255 [0.0604, 1.09]	0.140 [0.0300, 1.00]	0.193 [0.0300, 1.09]	
Non-quantified	5 (2.7%)	1 (0.7%)	6 (1.8%)	
PFUnDA				0.02

Mean (SD)	0.230 (0.125)	0.108 (0.0755)	0.175 (0.122)	
Median [Min, Max]	0.213 [0.0429, 0.680]	0.0900 [0.0300, 0.550]	0.135 [0.0300, 0.680]	
Non-quantified	13 (7.0%)	6 (4.1%)	19 (5.7%)	
PFDoDA				0.02
Mean (SD)	0.0624 (0.0158)	0.0346 (0.0273)	0.0412 (0.0274)	
Median [Min, Max]	0.0557 [0.0525, 0.0858]	0.0300 [0.0200, 0.120]	0.0300 [0.0200, 0.120]	
Non-quantified	181 (97.8%)	135 (91.2%)	316 (94.9%)	
PFBA				0.06
Mean (SD)	\	0.594 (1.22)	0.594 (1.22)	
Median [Min, Max]	\	0.0800 [0.0600, 5.20]	0.0800 [0.0600, 5.20]	
Non-quantified	185 (100%)	121 (81.8%)	306 (91.9%)	
PFPeA				0.01
Mean (SD)	\	0.0140 (0.00548)	0.0140 (0.00548)	
Median [Min, Max]	\	0.0100 [0.0100, 0.0200]	0.0100 [0.0100, 0.0200]	
Non-quantified	185 (100%)	143 (96.6%)	328 (98.5%)	
PFHxA				0.007
Mean (SD)	\	0.00867 (0.00153)	0.00867 (0.00153)	
Median [Min, Max]	\	0.00900 [0.0070, 0.010]	0.0090 [0.0070, 0.010]	
Non-quantified	185 (100%)	145 (98.0%)	330 (99.1%)	
PFTrDA				0.05
Mean (SD)	\	0.100 (\)	0.100 (\)	
Median [Min, Max]	\	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]	
Non-quantified	185 (100%)	147 (99.3%)	332 (99.7%)	
PFTeDA				0.05
Mean (SD)	\	0.150 (\)	0.150 (\)	
Median [Min, Max]	\	0.150 [0.150, 0.150]	0.150 [0.150, 0.150]	
Non-quantified	185 (100%)	147 (99.3%)	332 (99.7%)	
PFBS				0.01
Mean (SD)	\	0.132 (0.145)	0.132 (0.145)	

Median [Min, Max]	\	0.0750 [0.0100, 0.580]	0.0750 [0.0100, 0.580]	
Non-quantified	185 (100%)	122 (82.4%)	307 (92.2%)	
PFPeS				0.007
Mean (SD)	\	0.0147 (0.0165)	0.0147 (0.0165)	
Median [Min, Max]	\	0.0100 [0.00700, 0.0700]	0.0100 [0.00700, 0.0700]	
Non-quantified	185 (100%)	133 (89.9%)	318 (95.5%)	
PFHpS				0.007
Mean (SD)	\	0.0407 (0.0317)	0.0407 (0.0317)	
Median [Min, Max]	\	0.0300 [0.00800, 0.240]	0.0300 [0.00800, 0.240]	
Non-quantified	185 (100%)	1 (0.7%)	186 (55.9%)	
MeFOSAA				0.007
Mean (SD)	\	0.0399 (0.0682)	0.0399 (0.0682)	
Median [Min, Max]	\	0.0100 [0.00700, 0.310]	0.0100 [0.00700, 0.310]	
Non-quantified	185 (100%)	115 (77.7%)	300 (90.1%)	
N-EtFOSAA				0.007
Mean (SD)	\	0.0236 (0.00945)	0.0236 (0.00945)	
Median [Min, Max]	\	0.0272 [0.0100, 0.0300]	0.0272 [0.0100, 0.0300]	
Non-quantified	185 (100%)	144 (97.3%)	329 (98.8%)	
FOSA				0.01
Mean (SD)	\	0.0200 (0.0152)	0.0200 (0.0152)	
Median [Min, Max]	\	0.0100 [0.0100, 0.0700]	0.0100 [0.0100, 0.0700]	
Non-quantified	185 (100%)	114 (77.0%)	299 (89.8%)	
HFPO_DA				0.009
Mean (SD)	\	0.0621 (0.125)	0.0621 (0.125)	
Median [Min, Max]	\	0.0300 [0.00900, 0.780]	0.0300 [0.00900, 0.780]	
Non-quantified	185 (100%)	92 (62.2%)	277 (83.2%)	
9ClPF3OUdS				0.007
Mean (SD)	\	0.0198 (0.0161)	0.0198 (0.0161)	
Median [Min, Max]	\	0.0200 [0.00700, 0.110]	0.0200 [0.00700, 0.110]	

Non-quantified	185 (100%)	29 (19.6%)	214 (64.3%)	
MECPP				0.01
Mean (SD)	0.389 (1.85)	0.301 (0.213)	0.350 (1.38)	
Median [Min, Max]	0.190 [0.0600, 25.0]	0.230 [0.0900, 1.60]	0.210 [0.0600, 25.0]	
Non-quantified	3 (1.6%)	3 (2.0%)	6 (1.8%)	
MEHHP				0.01
Mean (SD)	0.0760 (0.348)	0.0628 (0.0208)	0.0703 (0.261)	
Median [Min, Max]	0.0400 [0.0200, 4.70]	0.0600 [0.0400, 0.130]	0.0500 [0.0200, 4.70]	
Non-quantified	5 (2.7%)	9 (6.1%)	14 (4.2%)	
MEOHP				
Mean (SD)	0.0473 (0.248)	0.0537 (0.0211)	0.0488 (0.215)	0.01
Median [Min, Max]	0.0200 [0.0100, 3.30]	0.0500 [0.0400, 0.170]	0.0300 [0.0100, 3.30]	
Non-quantified	10 (5.4%)	91 (61.5%)	101 (30.3%)	
MEHP				0.37
Mean (SD)	1.04 (1.65)	0.995 (0.418)	1.02 (1.29)	
Median [Min, Max]	0.775 [0.480, 21.0]	0.900 [0.680, 3.60]	0.850 [0.480, 21.0]	
Non-quantified	25 (13.5%)	35 (23.6%)	60 (18.0%)	
MnOP				0.008
Mean (SD)	0.0117 (0.00408)	\	0.0117 (0.00408)	
Median [Min, Max]	0.0100 [0.0100, 0.0200]	\	0.0100 [0.0100, 0.0200]	
Non-quantified	179 (96.8%)	148 (100%)	327 (98.2%)	
cxMiNCH				0.01
Mean (SD)	0.448 (1.85)	0.357 (1.51)	0.424 (1.76)	
Median [Min, Max]	0.0600 [0.0100, 16.0]	0.100 [0.0700, 12.0]	0.0800 [0.0100, 16.0]	
Non-quantified	10 (5.4%)	85 (57.4%)	95 (28.5%)	
ohMiNCH				0.02
Mean (SD)	0.334 (1.13)	0.101 (0.584)	0.195 (0.852)	
Median [Min, Max]	0.0500 [0.0200, 8.60]	0.0300 [0.0200, 5.90]	0.0300 [0.0200, 8.60]	
Non-quantified	116 (62.7%)	46 (31.1%)	162 (48.6%)	
oxoMiNCH				0.02

Mean (SD)	0.0909 (0.177)	0.0533 (0.127)	0.0804 (0.165)	
Median [Min, Max]	0.0600 [0.0300, 1.50]	0.0200 [0.0200, 0.680]	0.0500 [0.0200, 1.50]	
Non-quantified	115 (62.2%)	121 (81.8%)	236 (70.9%)	
cxMiNP				0.01
Mean (SD)	2.52 (7.03)	2.99 (6.62)	2.72 (6.85)	
Median [Min, Max]	0.610 [0.100, 70.0]	0.550 [0.160, 37.0]	0.580 [0.100, 70.0]	
Non-quantified	3 (1.6%)	5 (3.4%)	8 (2.4%)	
ohMiNP				0.007
Mean (SD)	1.01 (3.21)	0.918 (1.81)	0.976 (2.77)	
Median [Min, Max]	0.270 [0.0600, 33.0]	0.205 [0.0900, 11.0]	0.250 [0.0600, 33.0]	
Non-quantified	6 (3.2%)	40 (27.0%)	46 (13.8%)	
MEP				0.15
Mean (SD)	0.982 (1.09)	14.5 (160)	6.90 (106)	
Median [Min, Max]	0.740 [0.290, 10.0]	0.620 [0.290, 1900]	0.710 [0.290, 1900]	
Non-quantified	4 (2.2%)	7 (4.7%)	11 (3.3%)	
MCPH				0.005
Mean (SD)	0.0177 (0.0374)	0.0329 (0.00756)	0.0215 (0.0330)	
Median [Min, Max]	0.00800 [0.00700, 0.180]	0.0300 [0.0300, 0.0500]	0.00950 [0.00700, 0.180]	
Non-quantified	164 (88.6%)	141 (95.3%)	305 (91.6%)	
MiBP				1.7
Mean (SD)	3.17 (1.14)	4.30 (0.100)	3.23 (1.13)	
Median [Min, Max]	2.80 [2.50, 7.00]	4.30 [4.20, 4.40]	2.80 [2.50, 7.00]	
Non-quantified	139 (75.1%)	145 (98.0%)	284 (85.3%)	
MnBP				2.2
Mean (SD)	3.78 (0.806)	3.15 (0.105)	3.60 (0.740)	
Median [Min, Max]	3.30 [3.10, 5.20]	3.15 [3.00, 3.30]	3.30 [3.00, 5.20]	
Non-quantified	169 (91.4%)	142 (95.9%)	311 (93.4%)	
MOHiBP				0.01
Mean (SD)	0.0650 (0.0466)	0.0445 (0.0234)	0.0556 (0.0391)	
Median [Min, Max]	0.0500 [0.0200, 0.410]	0.0400 [0.0200, 0.160]	0.0500 [0.0200, 0.410]	
Non-quantified	13 (7.0%)	4 (2.7%)	17 (5.1%)	

MOHnBP				0.02
Mean (SD)	1.32 (5.97)	0.0386 (0.0124)	0.840 (4.76)	
Median [Min, Max]	0.0400 [0.0200, 42.0]	0.0400 [0.0200, 0.0800]	0.0400 [0.0200, 42.0]	
Non-quantified	69 (37.3%)	79 (53.4%)	148 (44.4%)	
MePAR				0.1
Mean (SD)	165.92 (678.72)	58.1 (39.08)	117.5 (506.5)	
Median [Min, Max]	7.9 [0.13, 6000]	54.0 [0.160, 230]	30.7 [0.130, 6000]	
Non-quantified	5 (2.7%)	1 (0.7%)	6 (1.8%)	
EtPAR				0.02
Mean (SD)	0.263 (0.387)	0.0843 (0.0940)	0.173 (0.295)	
Median [Min, Max]	0.125 [0.0300, 2.30]	0.0500 [0.0300, 0.630]	0.0600 [0.0300, 2.30]	
Non-quantified	95 (51.4%)	58 (39.2%)	153 (45.9%)	
iPrPAR				0.007
Mean (SD)	0.0854 (0.111)	0.0200 (\)	0.0843 (0.110)	
Median [Min, Max]	0.0300 [0.00900, 0.460]	0.0200 [0.0200, 0.0200]	0.0300 [0.00900, 0.460]	
Non-quantified	126 (68.1%)	147 (99.3%)	273 (82.0%)	
PrPAR				0.06
Mean (SD)	0.595 (1.47)	14.9 (15.0)	10.1 (14.0)	
Median [Min, Max]	0.175 [0.0800, 11.0]	12.0 [0.0700, 110]	6.90 [0.0700, 110]	
Non-quantified	111 (60.0%)	1 (0.7%)	112 (33.6%)	
iBuPAR				0.004
Mean (SD)	0.0304 (0.0449)	0.0220 (0.00447)	0.0290 (0.0410)	
Median [Min, Max]	0.00800 [0.00500, 0.160]	0.0200 [0.0200, 0.0300]	0.0100 [0.00500, 0.160]	
Non-quantified	160 (86.5%)	143 (96.6%)	303 (91.0%)	
BuPAR				0.04
Mean (SD)	0.512 (1.38)	0.0946 (0.0230)	0.316 (1.02)	
Median [Min, Max]	0.125 [0.0500, 9.40]	0.0900 [0.0700, 0.160]	0.0900 [0.0500, 9.40]	
Non-quantified	133 (71.9%)	102 (68.9%)	235 (70.6%)	
BPA				0.76
Mean (SD)	1.60 (0.382)	6.15 (9.23)	2.81 (4.77)	

Median [Min, Max]	1.60 [1.00, 2.30]	1.65 [1.30, 20.0]	1.60 [1.00, 20.0]	
Non-quantified	174 (94.1%)	144 (97.3%)	318 (95.5%)	
MBP				0.03
Mean (SD)	0.115 (0.0226)	\	0.115 (0.0226)	
Median [Min, Max]	0.115 [0.0900, 0.140]	\	0.115 [0.0900, 0.140]	
Non-quantified	179 (96.8%)	148 (100%)	327 (98.2%)	
BPF				0.02
Mean (SD)	0.117 (0.0758)	0.0800 (\)	0.115 (0.0741)	
Median [Min, Max]	0.100 [0.0400, 0.280]	0.0800 [0.0800, 0.0800]	0.100 [0.0400, 0.280]	
Non-quantified	168 (90.8%)	147 (99.3%)	315 (94.6%)	
BPS				0.04
Mean (SD)	0.0641 (0.0162)	0.111 (0.0616)	0.0873 (0.0505)	
Median [Min, Max]	0.0600 [0.0500, 0.110]	0.0950 [0.0700, 0.400]	0.0800 [0.0500, 0.400]	
Non-quantified	153 (82.7%)	116 (78.4%)	269 (80.8%)	
BPSIP				0.03
Mean (SD)	0.114 (0.106)	0.0704 (0.0606)	0.0823 (0.0773)	
Median [Min, Max]	0.0850 [0.0500, 0.470]	0.0500 [0.0500, 0.390]	0.0500 [0.0500, 0.470]	
Non-quantified	165 (89.2%)	95 (64.2%)	260 (78.1%)	
BPAF				0.02
Mean (SD)	0.0950 (0.106)	0.315 (0.00707)	0.205 (0.141)	
Median [Min, Max]	0.0950 [0.0200, 0.170]	0.315 [0.310, 0.320]	0.240 [0.0200, 0.320]	
Non-quantified	183 (98.9%)	146 (98.6%)	329 (98.8%)	
BPAP				0.03
Mean (SD)	\	0.168 (0.00500)	0.168 (0.00500)	
Median [Min, Max]	\	0.170 [0.160, 0.170]	0.170 [0.160, 0.170]	
Non-quantified	185 (100%)	144 (97.3%)	329 (98.8%)	
BPB				0.02
Mean (SD)	0.0400 (0.0141)	NA (NA)	0.0400 (0.0141)	
Median [Min, Max]	0.0400 [0.0300, 0.0500]	NA [NA, NA]	0.0400 [0.0300, 0.0500]	

Non-quantified	183 (98.9%)	148 (100%)	331 (99.4%)	
BPC2				0.01
Mean (SD)	\	0.0267 (0.00577)	0.0267 (0.00577)	
Median [Min, Max]	\	0.0300 [0.0200, 0.0300]	0.0300 [0.0200, 0.0300]	
Non-quantified	185 (100%)	145 (98.0%)	330 (99.1%)	

Table S4. Associations between phthalates and parabens concentrations and Ovarian Sensitivity Index, evaluated with individual regression models for each chemical^a

Plasticizer concentrations	Combined population (n=333) ^b		Sweden (n=185) ^c		Estonia (n=148)	
	Beta	95% CI	Beta	95% CI	Beta	95% CI
ΣDEHP						
Q1 [0.000331, 0.00306]	Ref	-	Ref	-	Ref	-
Q2 (0.00306, 0.00383]	-0.03	(-0.15, 0.09)	-0.10	(-0.25, 0.05)	-0.03	(-0.24, 0.19)
Q3 (0.00383, 0.00486]	-0.02	(-0.14, 0.11)	-0.10	(-0.27, 0.07)	-0.04	(-0.26, 0.18)
Q4 (0.00486, 0.0144]	-0.03	(-0.16, 0.10)	-0.22	(-0.41, -0.03)	0.01	(-0.21, 0.22)
MEP						
Q1 [0.285, 0.51]	Ref	-	Ref	-	Ref	-
Q2 (0.51, 0.71]	-0.06	(-0.18, 0.06)	0.07	(-0.12, 0.26)	-0.15	(-0.33, 0.03)
Q3 (0.71, 0.98]	0.0002	(-0.13, 0.13)	0.15	(-0.04, 0.35)	-0.08	(-0.31, 0.16)
Q4 (0.98, 12]	-0.02	(-0.14, 0.10)	0.02	(-0.19, 0.23)	0.01	(-0.16, 0.18)
cxMiNP						
Q1 [0.1, 0.3]	Ref	-	Ref	-	Ref	-
Q2 (0.3, 0.58]	-0.02	(-0.14, 0.10)	-0.07	(-0.25, 0.10)	-0.03	(-0.22, 0.15)
Q3 (0.58, 1.8]	-0.05	(-0.17, 0.07)	-0.05	(-0.22, 0.11)	-0.16	(-0.36, 0.05)
Q4 (1.8, 70]	-0.09	(-0.21, 0.04)	-0.08	(-0.26, 0.10)	-0.13	(-0.31, 0.06)
MOHiBP						
Q1 [0.02, 0.03]	Ref	-	Ref	-	Ref	-
Q2 (0.03, 0.05]	0.07	(-0.04, 0.18)	-0.08	(-0.25, 0.09)	0.15	(-0.004, 0.31)
Q3 (0.05, 0.06]	0.15	(0.01, 0.30)	0.08	(-0.12, 0.28)	0.15	(-0.09, 0.39)
Q4 (0.06, 0.41]	0.03	(-0.10, 0.16)	-0.12	(-0.30, 0.05)	0.17	(-0.06, 0.39)
Methylparaben						
Q1 [0.13, 5.3]	Ref	-	Ref	-	Ref	-
Q2 (5.3, 30]	0.08	(-0.04, 0.20)	0.08	(-0.06, 0.21)	-0.03	(-0.40, 0.34)
Q3 (30, 71]	-0.06	(-0.19, 0.08)	-0.04	(-0.27, 0.19)	-0.34	(-0.75, 0.08)
Q4 (71, 6000]	-0.02	(-0.15, 0.11)	-0.02	(-0.18, 0.13)	-0.29	(-0.80, 0.21)
cxMiNCH^d						
Q1 [0.01, 0.04]	Ref	-	Ref	-	Ref	-

Q2 (0.04, 0.08]	\	\	-0.02	(-0.17, 0.14)	\	\
Q3 (0.08, 0.15]	\	\	-0.11	(-0.28, 0.06)	\	\
Q4 (0.15, 16]	\	\	-0.15	(-0.30, 0.004)	\	\
Propylparaben ^e						
Q1 [0.07,0.35]	Ref	-	Ref	-	Ref	-
Q2 (0.35, 6.9]	\	\	\	\	-0.08	(-0.53, 0.36)
Q3 (6.9, 15]	\	\	\	\	0.14	(-0.39, 0.67)
Q4 (15, 110]	\	\	\	\	0.25	(-0.33, 0.83)

^aAdjusted for age, BMI, parity, and previous IVF.

^bLinear mixed model with cohort as a random intercept was used to account for inter-cohort heterogeneity.

^cFurther adjusted for infertility duration, smoking, fatty fish consumption, and PCP use.

^dHigh detection only in the Swedish population.

^eHigh detection only in the Estonian population.

Table S5. Associations between phthalates and parabens concentrations and Ovarian Sensitivity Index, evaluated with individual regression models for each chemical, each adjusting for potential mediators^a

Plasticizer concentrations	Combined population (n=333) ^b		Sweden (n=185) ^c		Estonia (n=148)	
	Beta	95% CI	Beta	95% CI	Beta	95% CI
ΣDEHP						
Q1 [0.000331, 0.00306]	Ref	-	Ref	-	Ref	-
Q2 (0.00306, 0.00383]	-0.02	(-0.13, 0.09)	-0.08	(-0.21, 0.06)	-0.05	(-0.29, 0.19)
Q3 (0.00383, 0.00486]	-0.05	(-0.17, 0.07)	-0.14	(-0.29, -0.001)	-0.01	(-0.25, 0.23)
Q4 (0.00486, 0.0144]	-0.10	(-0.22, 0.02)	-0.17	(-0.32, -0.02)	0.04	(-0.22, 0.30)
MEP						
Q1 [0.285, 0.51]	Ref	-	Ref	-	Ref	-
Q2 (0.51, 0.71]	0.03	(-0.09, 0.16)	0.10	(-0.07, 0.26)	-0.05	(-0.26, 0.17)
Q3 (0.71, 0.98]	0.07	(-0.06, 0.20)	0.10	(-0.07, 0.27)	-0.11	(-0.40, 0.19)
Q4 (0.98, 12]	0.05	(-0.08, 0.18)	0.04	(-0.13, 0.22)	0.07	(-0.15, 0.30)
cxMiNP						
Q1 [0.1, 0.3]	Ref	-	Ref	-	Ref	-
Q2 (0.3, 0.58]	0.05	(-0.07, 0.16)	-0.004	(-0.16, 0.15)	0.08	(-0.16, 0.33)
Q3 (0.58, 1.8]	0.01	(-0.10, 0.13)	-0.04	(-0.18, 0.11)	0.01	(-0.35, 0.36)
Q4 (1.8, 70]	-0.01	(-0.12, 0.10)	-0.05	(-0.20, 0.10)	-0.09	(-0.31, 0.12)
MOHiBP						
Q1 [0.02, 0.03]	Ref	-	Ref	-	Ref	-
Q2 (0.03, 0.05]	-0.02	(-0.14, 0.10)	-0.03	(-0.18, 0.12)	-0.09	(-0.28, 0.11)
Q3 (0.05, 0.06]	0.09	(-0.06, 0.23)	0.05	(-0.14, 0.24)	0.09	(-0.23, 0.42)
Q4 (0.06, 0.41]	-0.04	(-0.17, 0.08)	-0.08	(-0.23, 0.08)	0.13	(-0.10, 0.37)
Methylparaben						
Q1 [0.13, 5.3]	Ref	-	Ref	-	Ref	-
Q2 (5.3, 30]	-0.002	(-0.10, 0.10)	0.01	(-0.11, 0.13)	-0.03	(-0.49, 0.43)
Q3 (30, 71]	-0.07	(-0.21, 0.06)	-0.07	(-0.26, 0.13)	0.01	(-0.40, 0.43)
Q4 (71, 6000]	-0.01	(-0.12, 0.11)	-0.02	(-0.16, 0.13)	0.05	(-0.35, 0.45)
cxMiNCH^d						

Q1 [0.01, 0.04]	Ref	-	Ref	-	Ref	-
Q2 (0.04, 0.08]	\	\	0.01	(-0.13, 0.15)	\	\
Q3 (0.08, 0.15]	\	\	-0.05	(-0.20, 0.10)	\	\
Q4 (0.15, 16]	\	\	-0.07	(-0.22, 0.07)	\	\
Propylparaben ^e						
Q1 [0.07,0.35]	Ref	-	Ref	-	Ref	-
Q2 (0.35, 6.9]	\	\	\	\	0.20	(-0.28, 0.69)
Q3 (6.9, 15]	\	\	\	\	0.30	(-0.19, 0.79)
Q4 (15, 110]	\	\	\	\	0.31	(-0.19, 0.81)

^aAdjusted for age, BMI, parity, and previous IVF.

^bLinear mixed model with cohort as a random intercept was used to account for inter-cohort heterogeneity.

^cFurther adjusted for infertility duration, smoking, fatty fish consumption, and PCP use.

^dHigh detection only in the Swedish population.

^eHigh detection only in the Estonian population.

Table S6. Associations of phthalates and parabens concentrations with ovarian sensitivity index excluding infertility due to male causes, evaluated with individual regression models for each chemical^a

Plasticizer concentrations	Combined population (n=258) ^b			Sweden (n=141) ^c			Estonia (n=117)		
	n	Beta	95% CI	n	Beta	95% CI	n	Beta	95% CI
ΣDEHP									
Q1 [0.000331, 0.00306]	69	Ref	-	44	Ref	-	25	Ref	-
Q2 (0.00306, 0.00383]	59	-0.08	(-0.22, 0.05)	37	-0.10	(-0.27, 0.07)	22	-0.08	(-0.31, 0.14)
Q3 (0.00383, 0.00486]	63	-0.07	(-0.20, 0.06)	29	-0.10	(-0.29, 0.18)	34	-0.08	(-0.28, 0.13)
Q4 (0.00486, 0.0144]	63	-0.06	(-0.19, 0.07)	28	-0.21	(-0.41 -0.01)	35	0.03	(-0.17, 0.23)
MEP									
Q1 [0.285, 0.51]	64	Ref	-	22	Ref	-	42	Ref	-
Q2 (0.51, 0.71]	67	-0.05	(-0.18, 0.09)	42	0.05	(-0.17, 0.27)	25	-0.04	(-0.24, 0.16)
Q3 (0.71, 0.98]	58	-0.02	(-0.13, 0.17)	44	0.14	(-0.08, 0.36)	14	-0.07	(-0.31, 0.18)
Q4 (0.98, 12]	62	0.003	(-0.14, 0.10)	31	0.07	(-0.18, 0.31)	31	-0.01	(-0.19, 0.18)
cxMiNP									
Q1 [0.1, 0.3]	67	Ref	-	33	Ref	-	34	Ref	-
Q2 (0.3, 0.58]	59	0.05	(-0.09, 0.18)	35	-0.02	(-0.22, 0.18)	24	0.01	(-0.19, 0.21)
Q3 (0.58, 1.8]	65	-0.07	(-0.20, 0.06)	41	-0.14	(-0.33, 0.05)	24	-0.13	(-0.33, 0.07)
Q4 (1.8, 70]	60	-0.09	(-0.22, 0.05)	30	-0.06	(-0.26, 0.15)	30	-0.17	(-0.36, 0.02)
MOHiBP									
Q1 [0.02, 0.03]	64	Ref	-	18	Ref	-	46	Ref	-
Q2 (0.03, 0.05]	91	0.07	(-0.05, 0.20)	50	-0.04	(-0.25, 0.17)	41	0.14	(-0.02, 0.30)
Q3 (0.05, 0.06]	33	0.17	(0.005, 0.34)	21	0.03	(-0.22, 0.28)	12	0.29	(0.04, 0.53)
Q4 (0.06, 0.41]	59	0.08	(-0.06, 0.23)	44	-0.14	(-0.36, 0.08)	15	0.22	(-0.01, 0.46)
Methylparaben									
Q1 [0.13, 5.3]	57	Ref	-	48	Ref	-	9	Ref	-
Q2 (5.3, 30]	67	0.04	(-0.10, 0.18)	50	0.08	(-0.07, 0.24)	17	-0.12	(-0.43, 0.19)
Q3 (30, 71]	71	-0.09	(-0.25, 0.06)	12	0.03	(-0.24, 0.29)	59	-0.30	(-0.56, -0.04)
Q4 (71, 6000]	58	-0.02	(-0.17, 0.13)	27	-0.04	(-0.24, 0.16)	31	-0.09	(-0.36, 0.19)
cxMiNCH^d									

Q1 [0.01, 0.04]	\	Ref	-	58	Ref	-	\	Ref	-
Q2 (0.04, 0.08]	\	\	\	27	0.08	(-0.10, 0.26)	\	\	\
Q3 (0.08, 0.15]	\	\	\	17	-0.05	(-0.27, 0.17)	\	\	\
Q4 (0.15, 16]	\	\	\	33	-0.13	(-0.31, 0.05)	\	\	\
Propylparaben ^e									
Q1 [0.07, 0.35]	\	Ref	-	\	Ref	-	5	Ref	-
Q2 (0.35, 6.9]	\	\	\	\	\	\	27	-0.17	(-0.54, 0.20)
Q3 (6.9, 15]	\	\	\	\	\	\	46	-0.23	(-0.58, 0.12)
Q4 (15, 110]	\	\	\	\	\	\	38	-0.08	(-0.44, 0.27)

^aAdjusted for age, BMI, parity, and previous IVF.

^bFurther adjusted for cohort to account for structural differences between cohorts.

^cFurther adjusted for infertility duration, smoking, fatty fish consumption, and PCP use.

^dHigh detection only in the Swedish population.

^eHigh detection only in the Estonian population.

Table S7. Associations between PFAS concentrations and Ovarian Sensitivity Index, evaluated with individual regression models for each chemical^a

PFAS concentrations	Combined population (n=333) ^b		Sweden (n=185) ^c		Estonia (n=148)	
	Beta	95% CI	Beta	95% CI	Beta	95% CI
PFHxS						
Q1 [0.0137, 0.17]	Ref	-	Ref	-	Ref	-
Q2 (0.17, 0.33]	-0.05	(-0.17, 0.07)	0.15	(-0.13, 0.43)	-0.06	(-0.21, 0.08)
Q3 (0.33, 0.688]	0.01	(-0.12, 0.14)	0.16	(-0.11, 0.42)	-0.05	(-0.23, 0.14)
Q4 (0.688, 13]	0.01	(-0.13, 0.16)	0.19	(-0.08, 0.45)	0.10	(-0.28, 0.48)
PFOA						
Q1 [0.169, 0.604]	Ref	-	Ref	-	Ref	-
Q2 (0.604, 0.85]	0.07	(-0.05, 0.18)	0.12	(-0.10, 0.34)	0.07	(-0.08, 0.21)
Q3 (0.85, 1.33]	0.11	(-0.02, 0.23)	0.32	(0.11, 0.53)	0.03	(-0.15, 0.20)
Q4 (1.33, 8.05]	0.05	(-0.09, 0.19)	0.25	(0.03, 0.46)	-0.15	(-0.39, 0.10)
PFOS						
Q1 [0.171, 1.84]	Ref	-	Ref	-	Ref	-
Q2 (1.84, 2.69]	-0.02	(-0.14, 0.10)	0.24	(0.002, 0.48)	-0.14	(-0.30, 0.03)
Q3 (2.69, 4.34]	-0.10	(-0.23, 0.03)	0.14	(-0.09, 0.37)	-0.10	(-0.29, 0.09)
Q4 (4.34, 15.1]	0.03	(-0.10, 0.16)	0.32	(0.08, 0.55)	-0.07	(-0.27, 0.13)
PFNA						
Q1 [0.0204, 0.3]	Ref	-	Ref	-	Ref	-
Q2 (0.3, 0.441]	0.07	(-0.05, 0.19)	0.22	(0.02, 0.42)	-0.02	(-0.18, 0.14)
Q3 (0.441, 0.642]	-0.04	(-0.15, 0.08)	0.23	(0.04, 0.42)	-0.19	(-0.36, -0.02)
Q4 (0.642, 3.16]	0.03	(-0.10, 0.15)	0.17	(-0.03, 0.37)	-0.03	(-0.22, 0.16)
PFUnDA						
Q1 [0.03, 0.08]	Ref	-	Ref	-	Ref	-
Q2 (0.08, 0.135]	-0.15	(-0.27, -0.03)	0.05	(-0.19, 0.29)	-0.22	(-0.36, -0.07)
Q3 (0.135, 0.245]	-0.03	(-0.16, 0.09)	0.16	(-0.06, 0.38)	-0.08	(-0.25, 0.09)
Q4 (0.245, 0.68]	-0.06	(-0.21, 0.08)	0.13	(-0.09, 0.34)	-0.19	(-0.50, 0.13)
PFDA						
Q1 [0.03, 0.132]	Ref	-	Ref	-	Ref	-
Q2 (0.132, 0.193]	-0.09	(-0.21, 0.03)	0.06	(-0.18, 0.30)	-0.13	(-0.29, 0.02)

Q3 (0.193, 0.31]	-0.03	(-0.15, 0.10)	0.23	(0.0002, 0.46)	-0.14	(-0.32, 0.03)
Q4 (0.31, 1.09]	-0.07	(-0.20, 0.06)	0.07	(-0.16, 0.31)	-0.02	(-0.23, 0.18)

^aAdjusted for age, BMI, parity, and previous IVF.

^bFurther adjusted for cohort to account for structural differences between cohorts.

^cFurther adjusted for infertility duration, smoking, fatty fish consumption, and PCP use.

Table S8. Associations between PFAS concentrations and Ovarian Sensitivity Index, evaluated with a single regression model, mutually adjusting for all chemicals and adjusting for potential mediators^a

PFAS concentrations	Overall (n=333) ^b		Sweden (n=185) ^c		Estonia (n=148)	
	Beta	95% CI	Beta	95% CI	Beta	95% CI
PFHxS						
Q1 [0.0137, 0.17]	Ref	-	Ref	-	Ref	-
Q2 (0.17, 0.33]	0.11	(-0.09, 0.32)	0.01	(-0.28, 0.29)	0.36	(-0.20, 0.93)
Q3 (0.33, 0.688]	0.12	(-0.10, 0.33)	0.01	(-0.26, 0.29)	0.20	(-0.70, 1.11)
Q4 (0.688, 13]	0.10	(-0.12, 0.32)	0.05	(-0.23, 0.33)	0.35	(-0.40, 1.09)
PFOA						
Q1 [0.169, 0.604]	Ref	-	Ref	-	Ref	-
Q2 (0.604, 0.85]	-0.09	(-0.25, 0.08)	-0.02	(-0.24, 0.20)	0.24	(-0.66, 1.15)
Q3 (0.85, 1.33]	-0.01	(-0.17, 0.16)	0.09	(-0.12, 0.30)	0.10	(-0.92, 1.11)
Q4 (1.33, 8.05]	-0.06	(-0.24, 0.13)	0.003	(-0.23, 0.23)	0.16	(-0.84, 1.17)
PFOS						
Q1 [0.171, 1.84]	Ref	-	Ref	-	Ref	-
Q2 (1.84, 2.69]	-0.06	(-0.28, 0.15)	-0.01	(-0.29, 0.28)	0.12	(-1.15, 1.39)
Q3 (2.69, 4.34]	-0.14	(-0.36, 0.08)	-0.13	(-0.41, 0.16)	0.18	(-1.57, 1.93)
Q4 (4.34, 15.1]	0.01	(-0.23, 0.25)	0.04	(-0.26, 0.34)	-0.16	(-1.42, 1.10)
PFNA						
Q1 [0.0204, 0.3]	Ref	-	Ref	-	Ref	-
Q2 (0.3, 0.441]	-0.05	(-0.23, 0.13)	0.07	(-0.16, 0.30)	-0.50	(-2.10, 1.09)
Q3 (0.441, 0.642]	0.07	(-0.15, 0.29)	0.17	(-0.13, 0.47)	-0.26	(-1.64, 1.12)
Q4 (0.642, 3.16]	0.02	(-0.24, 0.28)	0.13	(-0.21, 0.47)	0.05	(-1.59, 1.69)
PFUnDA						
Q1 [0.03, 0.08]	Ref	-	Ref	-	Ref	-
Q2 (0.08, 0.135]	-0.04	(-0.22, 0.14)	0.04	(-0.18, 0.25)	-0.41	(-2.21, 1.40)
Q3 (0.135, 0.245]	0.06	(-0.14, 0.26)	-0.01	(-0.24, 0.22)	-0.19	(-2.81, 2.43)
Q4 (0.245, 0.68]	0.13	(-0.12, 0.37)	0.09	(-0.19, 0.36)	-0.51	(-4.14, 3.13)
PFDA						
Q1 [0.03, 0.132]	Ref	-	Ref	-	Ref	-
Q2 (0.132, 0.193]	0.02	(-0.18, 0.22)	0.02	(-0.25, 0.29)	0.53	(-1.68, 2.73)

Q3 (0.193, 0.31]	-0.08	(-0.33, 0.16)	0.04	(-0.28, 0.36)	0.06	(-2.01, 2.13)
Q4 (0.31, 1.09]	-0.21	(-0.48, 0.08)	-0.14	(-0.48, 0.20)	0.40	(-2.40, 3.20)

^aAdjusted for age, BMI, parity, and previous IVF.

^bFurther adjusted for cohort to account for structural differences between cohorts.

^cFurther adjusted for infertility duration, smoking, fatty fish consumption, and PCP use.

Figures S3

Overall mixture effect on clinical pregnancy (CP) and live birth (LB), adjusted for age, BMI, parity, previous IVF, estimated from Hierarchical Bayesian Kernel Machine Regression (BKMR) in a subsample with complete information (n=234 from fresh transfers, and n=258 from fresh/frozen transfers).

